

Project Handover & Completion Report

Delivered scope, acceptance status, and formal transfer of the CareBridge healthcare referral and settlement platform

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1. Executive Summary

CareBridge is a healthcare referral and settlement platform that connects referring consultants with hospitals and diagnostic laboratories, and manages the commercial relationship between them end to end. This report closes the development engagement: it states what was built, confirms the current status of each delivered module, records what remains outside the delivered scope, and sets out the terms on which the system is transferred to the client.

The platform is complete and deployed. It comprises a React single-page application served from Vercel and a Node.js REST and real-time API served from Railway, backed by MongoDB. Four user roles are supported — consultant, hospital, laboratory, and platform administrator — each with its own portal, permissions, and workflows. The commercial engine that splits patient bills between the facility, the referring consultant, and the platform is implemented as a single tested module and covers both the original nested commission model and the additive per-doctor model adopted later in the engagement.

1.1 Delivery at a Glance

Measure	Value
Delivered modules	9 functional modules across 4 user roles
REST API endpoints	130+ endpoints across 14 routers, all versioned under /v1
Data model	17 MongoDB collections
Application screens	48 routed screens
Source code	Approximately 31,400 lines of application code (excluding dependencies)
Automated tests	40 tests across 5 suites — all passing
Development period	3 May 2026 to 8 July 2026
Deployment	Live — frontend on Vercel, backend and scheduler on Railway, database on MongoDB

1.2 Statement of Completion

All modules described in Section 3 of this report are implemented, deployed, and operational. The automated test suite passes in full. Requirements raised after the final code change of this engagement are not part of this delivery and are recorded in Section 6 as outstanding scope for a subsequent phase.

READ THIS SECTION ALONGSIDE SECTION 6

This report certifies the delivered scope as complete. It does not certify that every requirement ever discussed during the engagement has been built. Section 6 records the scope that remains outstanding, including requirements raised after the final code change. Please review Section 6 before signing the acceptance block in Section 8.

2. Project Overview

2.1 Purpose of the System

Patient referral in the private healthcare sector is typically informal — a consultant telephones a hospital, the patient is sent, and the commercial arrangement between the parties is settled privately and inconsistently. CareBridge formalises this into a tracked, auditable process. A consultant creates a structured referral; the platform recommends suitable hospitals based on clinical fit, bed availability, distance, and past performance; the hospital responds within a deadline; the patient is admitted, treated, and billed; and the resulting money is split between the parties according to terms held in the system rather than in individual memory.

2.2 Business Objectives Addressed

Objective	How the delivered system addresses it
Route referrals to the right facility	A weighted scoring engine ranks hospitals against each referral on six configurable factors, hard-filtering any hospital that lacks the required department or a free bed in the required ward.
Guarantee a response to the referring doctor	Every referral carries a response deadline set by clinical urgency. A scheduler checks every minute and automatically escalates unanswered referrals to the next hospital in the ranked list, or — for emergencies — to the nearest alternative within 20 km.
Make the commercial split explicit and consistent	A single commission engine computes every bill split. Terms are held per platform, per facility, and per consultant-facility pairing, and the terms applied are snapshotted onto each payout record so historical calculations remain reproducible after rates change.
Provide an auditable settlement cycle	Hospitals and laboratories bundle billed cases into periodic settlements, pay the platform, and upload receipts. Administrators verify each receipt before consultant payouts are released, and consultants confirm receipt to close the cycle.
Protect patient information	Patient national identity numbers are encrypted at the field level before storage. Access to every screen and endpoint is controlled by role, and administrative actions are recorded in an audit log.
Give the platform operator control	Administrators configure commercial terms, matching-engine weights, the department catalogue, and platform branding from the admin console, without code changes or redeployment.

2.3 Users of the System

Role	Description	Primary responsibility
Consultant	Referring doctors in private practice or clinics	Create and track referrals; receive commission

Role	Description	Primary responsibility
Hospital	Hospital administrators and their staff accounts	Accept referrals, admit and bill patients, settle with the platform
Laboratory	Diagnostic laboratory administrators and staff	Accept test referrals, report results, bill and settle
Administrator	CareBridge platform operations staff	Approve registrations, set commercial terms, verify settlements, oversee the platform

3. Delivered Scope

The following modules constitute the delivered system. Every item marked Implemented is present in the deployed codebase and operational.

3.1 Module 1 — Identity, Registration, and Access Control

Status	Capability
Implemented	Self-service registration for consultants, hospitals, and laboratories, each with role-specific fields
Implemented	Mandatory upload of verification documents (professional registration certificates and identity documents) at registration
Implemented	Mandatory acceptance of the standard operating procedure and privacy policy before submission
Implemented	Six-digit email one-time-password verification, with resend and cooldown
Implemented	Administrator approval gate — accounts are created pending and cannot sign in until reviewed and activated
Implemented	Rejection of an application with a stated reason, communicated to the applicant
Implemented	Password reset by emailed single-use time-limited link; administrator-initiated password reset
Implemented	Session management using a short-lived access token with rolling refresh, refreshed transparently in the browser
Implemented	Role-based route and endpoint authorisation across all four roles
Implemented	Sub-account creation so hospitals, laboratories, and the platform can add team members with their own logins
Implemented	Account suspension and reactivation without loss of history

3.2 Module 2 — Referral Creation and Intelligent Matching

Status	Capability
Implemented	Four-step guided referral intake covering patient demographics, clinical assessment, preferences, and hospital selection
Implemented	Age entry in years, months, or days with automatic derivation of date of birth, supporting paediatric and neonatal referrals
Implemented	Field-level encryption of the patient national identity number
Implemented	Automatic department suggestion from free-text symptoms using an administrator-maintained keyword catalogue

Status	Capability
Implemented	Clinical urgency classification (emergency, urgent, routine) driving the hospital response deadline
Implemented	Attachment upload for prior reports, scans, and images
Implemented	Weighted hospital scoring on six factors: department match, bed availability, distance, cost fit, response-time history, and consultant preference
Implemented	Hard filtering of hospitals lacking the required department or a free bed in the required ward
Implemented	Selection of a specific department and named doctor at the receiving hospital
Implemented	Submission to multiple hospitals as independently tracked referrals
Implemented	Consultant favourites and automatic learning from referral history, both feeding the preference factor
Implemented	Unique sequential referral codes allocated atomically

3.3 Module 3 — Referral Lifecycle and Escalation

Status	Capability
Implemented	Hospital inbox with live per-referral countdown to the response deadline
Implemented	Dedicated emergency triage board filtered to unanswered emergency referrals
Implemented	Accept with mandatory department assignment; reject with mandatory reason
Implemented	Response deadlines by urgency — 15 minutes for emergencies, 2 hours for urgent, 24 hours for routine
Implemented	Automatic escalation every minute of any referral past its deadline to the next ranked hospital, with a fresh deadline
Implemented	Geographic fallback for emergencies — escalation to the nearest active alternative hospital within 20 km when the ranked list is exhausted
Implemented	Automatic rejection with an explanatory reason when no hospital remains
Implemented	Performance penalty applied to a hospital that allows a referral to escalate, feeding back into its future ranking
Implemented	Shared clinical notes timeline, distinguishing nursing and consultant entries, visible to the referring consultant
Implemented	Full referral status lifecycle: pending, accepted, rejected, admitted, closed

3.4 Module 4 — Admissions, Beds, and Hospital Operations

Status	Capability
Implemented	Admission of an accepted referral with ward, room, bed, department, and treating doctor
Implemented	Line-item billing on discharge with automatic totalling, payment method, reference, and attached patient bill
Implemented	Bed inventory across ten ward types with quick adjustment and direct editing
Implemented	Department management determining which referrals the hospital is eligible to receive
Implemented	Hospital doctor register with specialty, registration number, consultation fee, and availability toggle
Implemented	Financial ledger of billed amounts, platform charges, and net position, exportable to CSV
Implemented	Hospital-level portal branding — logo and primary colour applied for that hospital's users
Implemented	Hospital team management with creation-hierarchy protection

3.5 Module 5 — Laboratory Module

Status	Capability
Implemented	Laboratory registration including an initial test catalogue with prices and turnaround times
Implemented	Laboratory referral creation by consultants, with an ordered test list and an optional patient discount capped per consultant
Implemented	Laboratory suggestion and direct selection by name
Implemented	Acceptance requiring a committed expected report date and time
Implemented	Report file upload with immediate notification to the consultant
Implemented	Per-test billing pre-filled from the laboratory's catalogue, with test descriptions locked to the consultant's order
Implemented	Re-referral of a rejected case to an alternative laboratory without re-entry of patient details
Implemented	Independent laboratory settlement cycle mirroring the hospital cycle
Implemented	Laboratory test catalogue maintenance and team management

3.6 Module 6 — Commission and Settlement Engine

Status	Capability
Implemented	Single commission engine used by every calculation path, eliminating the duplicated bill-splitting logic identified during design

Status	Capability
Implemented	Nested (legacy) commission model — platform takes a percentage of the bill, the consultant takes a percentage of that
Implemented	Additive (per-doctor) commission model — consultant fee and platform charge computed independently and summed
Implemented	Percentage or fixed amount, selectable independently for the consultant fee and the platform charge
Implemented	Per-referral fixed charges for hospitals and per-test fixed charges for laboratories
Implemented	Three-tier terms resolution: consultant-facility override, then facility terms, then platform defaults
Implemented	Snapshotting of the terms applied onto every payout record, so historical figures remain reproducible after rates change
Implemented	Mixed settlements — a single settlement period may contain consultants on different models, types, and rates
Implemented	Five-stage settlement state machine for both hospitals and laboratories, from pending payment through to completed
Implemented	Receipt upload by the facility, verification or rejection with reason by the administrator, payout release, and consultant confirmation of receipt
Implemented	Consultant earnings dashboard, withdrawal request with a minimum threshold, and exportable earnings statement
Implemented	All monetary values stored as integer paisa, avoiding floating-point rounding error

3.7 Module 7 — Platform Administration

Status	Capability
Implemented	Registration approval queue with document review
Implemented	Consultant, hospital, and laboratory registries with profile editing, status control, forced password reset, and export
Implemented	Commercial terms configuration at platform, facility, and consultant-facility level
Implemented	Matching-engine weight configuration with enforced validation that weights total exactly 100
Implemented	Department and symptom-keyword catalogue maintenance
Implemented	Platform-wide referral oversight, including correction of any referral and a three-stage confirmed deletion
Implemented	Cross-hospital bed inventory oversight and correction
Implemented	Settlement verification and payout release for both hospitals and laboratories

Status	Capability
Implemented	Financial defaults configuration, including wallet threshold and initial hold
Implemented	Platform white-labelling — name, logo, favicon, and colour scheme
Implemented	Searchable and exportable audit log of administrative actions
Implemented	Administrator team management

3.8 Module 8 — Notifications and Real-Time Updates

Status	Capability
Implemented	Eighteen notification event types spanning registration, referral, admission, billing, settlement, and account events
Implemented	In-application notification centre with unread counter and live delivery
Implemented	Email notification channel
Implemented	Live screen updates over a persistent connection — dashboards, inboxes, and bed counts refresh without user action
Implemented	Notification fan-out to facility owners and all their team members
Implemented	Removal of personal data from stored notification payloads
Partial	WhatsApp notification channel — the integration is implemented and documented, but requires the client to complete Meta Business template approval before live messages are delivered. See the Deployment and Operations Guide.

3.9 Module 9 — Documents, Exports, and Payments

Status	Capability
Implemented	Server-generated PDF records for referrals, consultant rosters, hospital records, laboratory records, and profile files, with attachments merged into the output
Implemented	Consultant earnings statement and laboratory earnings export generated in the browser
Implemented	Hospital financial ledger export to CSV
Implemented	Audit log export
Implemented	Document upload to cloud storage with type and size validation
Partial	JazzCash online payment — the integration including request signing and callback verification is implemented, but the platform currently operates on the manual receipt-based settlement cycle. Activation requires live merchant credentials from the client. See the Deployment and Operations Guide.

4. Technical Delivery

4.1 Technology Stack

Layer	Technology	Rationale
Frontend	React 19, Vite 8, Tailwind CSS 4, React Router 7, TanStack Query 5	Component model suited to role-specific portals; server-state caching reduces redundant requests; utility styling supports runtime white-labelling.
Backend	Node.js 20+, Express 5	Single language across the stack; mature ecosystem for the required integrations.
Database	MongoDB with Mongoose 9	Document model fits the varying shape of referral, admission, and settlement records; native geospatial indexing supports distance-based matching and emergency fallback.
Real-time	Socket.IO 4	Push updates to inboxes and dashboards; room-based addressing targets individual users, roles, and facilities.
Scheduling	node-cron	Runs the per-minute referral escalation check in the API process.
File storage	Cloudinary	Offloads document and report storage from the application server.
Email	Resend	Transactional delivery for verification codes, password resets, and notifications.
Messaging	Meta WhatsApp Cloud API	Notification channel appropriate to the user base.
Payments	JazzCash	Local payment gateway integration.
Hosting	Vercel (frontend), Railway (backend)	Managed platforms with automatic deployment from the repository.

4.2 Architecture

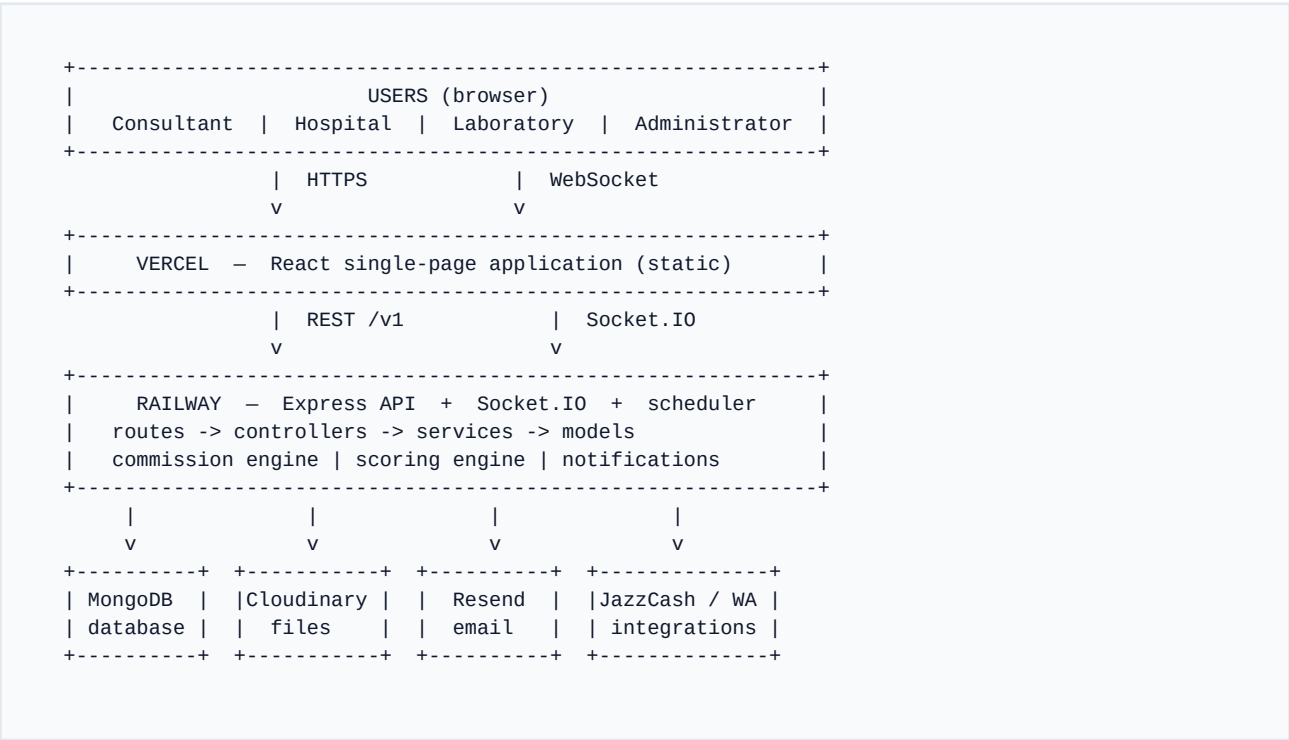


Figure 1: Deployed system architecture

4.3 Deployment Position

Component	Platform	Deployment method
Frontend application	Vercel	Built with Vite and served as static assets, with single-page routing rewrites and security response headers configured.
Backend API and scheduler	Railway	Run directly from source with automatic restart on failure.
Database	MongoDB	Managed instance reached over an authenticated connection string.
File storage	Cloudinary	Managed service, credentials held in backend configuration.

5. Quality Assurance

5.1 Automated Test Results

The automated suite runs against the backend business logic. It was executed in full at the time of this report with the following result.

Suite	Tests	Result	Coverage
Commission engine	17	Pass	Both commission models, percentage and fixed types for hospitals and laboratories, facility and consultant overrides, the invariant that the total deduction equals the consultant fee plus the platform charge, zero-bill handling, and the case where a fixed charge exceeds a small bill.
Scoring engine	6	Pass	Score composition on a full match, rejection when the department is absent or no bed is free, emergency intensive-care requirement, the 100-point cap, and distance calculation.
PDF export service	12	Pass	Validity of all eight document builders, single-page output for short records, image attachments embedded as pages, and PDF attachments merged page for page.
Field encryption	4	Pass	Encryption and decryption round trip, distinct output for repeated encryption of the same value, and safe handling of empty and unencrypted values.
Account hierarchy	1	Pass	Creation-chain ancestry check used to authorise team member deletion.
Total	40	Pass	All suites passing, no failures, no skipped tests.

WHAT THE AUTOMATED SUITE DOES AND DOES NOT COVER

The suite covers the calculation-heavy logic where a defect would be financially material and hardest to detect by inspection — bill splitting, hospital scoring, encryption, and document generation. It does not include HTTP-level integration tests of the controllers and routes, the escalation scheduler, or the settlement state machines; those areas were validated by manual testing. This is recorded as a recommendation in the Test Plan and QA Report.

5.2 Manual Verification

Each role's end-to-end workflow was exercised manually against the deployed environment: registration through to approval and first sign-in; referral creation through hospital acceptance, admission, billing, and closure; laboratory referral through reporting and billing; and a full settlement cycle through to consultant confirmation of receipt. Detail is recorded in the Test Plan and QA Report.

6. Outstanding Scope and Known Limitations

This section is a complete and deliberate record of what the delivered system does not do. It exists so that the client can make an informed acceptance decision and plan any subsequent phase accurately.

6.1 Client Requirements Not Included in This Delivery

Requirements received after the final code change of this engagement are outside the scope of this delivery and are tracked separately. They are not implemented in the delivered system and should be scoped and quoted as a subsequent phase.

NEW SCOPE, NOT DEFECTS

Scope raised after the close of the engagement is not a defect in what was delivered. Where such requirements affect the data model, the API authorisation layer, or the user interface, they should be estimated as new work rather than treated as remedial.

6.2 Technical Limitations of the Delivered System

The following are characteristics of the system as delivered that the client should be aware of. Each is described with its practical consequence. Recommended remediation is given in the Security and Compliance document and the Maintenance and Support Guide.

Limitation	Practical consequence
Administrator role bypasses role checks	The authorisation layer grants administrators access to every endpoint regardless of the role stated for that endpoint. This is intentional for platform oversight, but it means an administrator account is unrestricted and should be issued sparingly and protected accordingly.
The file upload endpoint does not require authentication	Files can be uploaded without signing in. Uploads are constrained to images and PDF documents of 5 MB or less, and an uploaded file is only meaningful once attached to a record by an authenticated user, but the endpoint should be placed behind authentication. This is the highest-priority item in the recommendations.
One-time passwords are held in application memory	Verification codes do not survive a restart of the backend, and the backend cannot be scaled to more than one instance without codes failing intermittently. Moving this store to a shared cache is required before horizontal scaling.
Rate limiting applies only to authentication endpoints	Registration, sign-in, and password reset are throttled. The remainder of the API is not, leaving it without protection against automated abuse.
Patient identity encryption is unauthenticated	The encryption in use provides confidentiality but not tamper detection — modification of stored ciphertext would not be detected on decryption. Upgrading to an authenticated encryption mode is recommended.
Cross-origin policy defaults to permissive when unset	If the allowed-origins setting is left blank, the API accepts requests from any origin. It must be set explicitly in the production environment.

Limitation	Practical consequence
A superseded wallet crediting path remains in the codebase	An earlier automatic wallet-crediting module remains present but is not connected to the current settlement flow. It duplicates commission arithmetic outside the tested engine and should be removed to prevent future accidental use.
No integration or end-to-end automated tests	Controllers, routes, the escalation scheduler, and the settlement state machines are covered by manual testing only, so regressions in those areas will not be caught automatically.
Two built screens are not reachable	A WhatsApp broadcast console and a public landing page exist in the codebase but are not linked into the application. They can be enabled with a small change if wanted.

6.3 Client Actions Required to Activate Optional Features

Feature	Action required by the client
WhatsApp notifications	Complete Meta Business verification and obtain approval for the two message templates, then supply the resulting credentials for configuration. The setup runbook is included in the handover package.
JazzCash online payments	Supply live merchant credentials. The platform currently operates the manual receipt-based settlement cycle, which is fully functional without this.
Custom email sending domain	Verify the sending domain with the email provider so notifications are sent from a CareBridge address rather than the provider default.

7. Handover Package

7.1 Documentation Delivered

Reference	Document	Purpose
CB-PHR-2026-001	Project Handover & Completion Report	This document — delivered scope, status, and formal transfer.
CB-SRS-2026-002	Software Requirements Specification	Functional and non-functional requirements, each traced to its implementation status.
CB-SAD-2026-003	System Architecture & Design Document	Architecture, module design, data model, and the commission and scoring algorithms.
CB-API-2026-004	API Reference	Complete endpoint catalogue, authentication model, error handling, and real-time events.
CB-DOG-2026-005	Deployment & Operations Guide	Deployment procedures, configuration reference, backup and restore, and operational runbooks.
CB-QA-2026-006	Test Plan & QA Report	Test strategy, automated results, acceptance scenarios by role, and defect position.
CB-UM-2026-007	User Manual	Operating instructions for all four roles.
CB-SEC-2026-008	Security & Compliance Document	Security architecture, access control matrix, data protection, and assessment of the delivered security model.
CB-MSG-2026-009	Maintenance & Support Guide	Codebase orientation, routine maintenance, troubleshooting, and the recommended roadmap.

7.2 Assets Transferred

- Source code** — the complete repository including frontend, backend, tests, and the documentation generation scripts used to produce this package.
- Configuration templates** — a documented environment file listing every setting the system reads, with guidance on each.
- Operational scripts** — administrator seeding, data maintenance, and verification utilities.
- Design records** — the internal design documents tracing the evolution of the commission model through its three generations, and the WhatsApp production setup runbook.
- Deployment configuration** — the hosting configuration files for both platforms, committed to the repository.

7.3 Credentials and Accounts

CREDENTIALS ARE TRANSFERRED SEPARATELY

Access credentials for the hosting platforms, the database, and the third-party services are not contained in this document or any other document in this package. They are transferred by a separate secure channel. On receipt, the client should rotate every shared secret — in particular the authentication signing keys and the database password — so that the delivered system operates on secrets known only to the client. The procedure is set out in the Deployment and Operations Guide.

8. Acceptance and Sign-Off

8.1 Basis of Acceptance

By signing below, the client acknowledges that the modules listed in Section 3 have been delivered and demonstrated; that the outstanding scope and technical limitations recorded in Section 6 have been read and understood, including requirements raised after the close of the engagement that are not implemented; and that the documentation and assets listed in Section 7 have been received.

8.2 Recommended Immediate Actions

The following should be completed by the client within the first week of taking ownership. Each is explained in the Deployment and Operations Guide.

- 1. **Rotate all shared secrets** — the two authentication signing keys, the database password, and every third-party service key.
- 2. Set the allowed-origins configuration explicitly to the production frontend address, so the API is not open to arbitrary origins.
- 3. **Place the file upload endpoint behind authentication** — the single highest-priority security action.
- 4. Confirm that automated database backups are enabled and verify a restore into a test environment.
- 5. Change the seeded administrator password and create named administrator accounts for each operator, so that audit log entries identify individuals.
- 6. Review the outstanding scope in Section 6 and decide which items to commission as a subsequent phase.

8.3 Signatures

This report is issued in the version and on the date shown on the cover page.

For the Client
(CareBridge Health)

Name

Signature

Date

For the Development Team

Name

Signature

Date